

NBA Lineup prediction

[Start Assignment](#)

- Due Wednesday by 11:59a.m.
- Points 25
- Submitting a file upload
- File Types ppt and pptx

Project Title: **NBA Lineup Prediction for Optimized Team Performance**

Objective:

The goal of this project is to design and develop a machine learning model that predicts the optimal fifth player for a home team in an NBA game, given partial lineup data and other game-related features. The model should maximize the home team's overall performance.

Project Details:

Dataset:

- You have been provided with a dataset containing lineups of all NBA games from 2007 to 2015.
- The dataset includes game-related features, player statistics, team compositions, and game outcomes.

Feature Restrictions:

- Not all features from the dataset are permitted for use in the model.
- A separate file will be provided that specifies which features are allowed. You must strictly adhere to this file when selecting input features for their model.

Training and Testing:

- **Training Data:** Historical data of NBA games with full lineups and allowed features, including the performance outcomes of teams.
- **Test Data:** Each test sample will be selected from game segments where the home team's performance (outcome=1) is better than the away team's performance. Each test sample will include:
 1. The lineup of four players from the home team.
 2. The lineup of five players from the away team.
 3. Other relevant game-related features.

Task:

- Build a machine learning model using the provided training data.
- Predict the fifth player for the home team based on the test data.
- The predicted player should optimize the home team's performance based on historical patterns and features.

Guidelines and Constraints:

1. **Feature Usage:** Only the features specified in the provided file may be used for model training and testing.
2. **Evaluation Metric:** The model's performance will be evaluated based on:
 - The predicted player's historical contribution to team success in similar scenarios.
 - Team performance metrics are derived from the predicted lineup.
3. **Model Explainability:** You must include an explanation of how the model makes predictions, particularly the reasoning behind selecting a specific player as the fifth member.
4. **Data Preprocessing:** You are required to perform:
 - Cleaning of the dataset (handling missing values, standardizing formats, etc.).
 - Feature selection based on the provided allowed feature file.
5. **Prediction Constraints:**
 - The recommended fifth player must be chosen from the roster of eligible players in the dataset.
 - Models must avoid recommending players who are absent due to injury or other constraints (use provided metadata).
6. **Code and Documentation Repository:**
 - All code and the final project report must be maintained in a GitHub repository.
 - The repository must be well-organized with a `README` file that explains how to run the code and interpret the results.

Deliverables:

1. **GitHub Repository:**
 - The complete project code.
 - The project report (in GitHub and PDF format) documents the approach, methods, and results.
 - A `README` file detailing:
 - Project objectives.
 - Instructions for setting up and running the code.
 - An overview of the results.
2. **Model Implementation:**
 - The final machine learning model with properly documented code.
 - A clear explanation of the model's architecture and choice of algorithm(s).
3. **Prediction Outputs:**
 - Submit predictions for the test data (fifth player recommendations).
 - Format: `Game_ID, Home_Team, Fifth_Player`
4. **Report:**
 - Detailed explanation of the data preprocessing steps, feature selection, and model evaluation.
 - Include visualizations and analysis of the results.
5. **Presentation:**

- A 10-minute presentation summarizing the project approach, challenges faced, and findings.
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Evaluation Criteria:

1. **Model Accuracy (40%):** Effectiveness of the model in predicting the optimal fifth player.
2. **Adherence to Feature Constraints (20%):** Proper use of only the allowed features.
3. **Code Quality and GitHub Usage (10%):** Organization, clarity, and thoroughness of the GitHub repository.
4. **Presentation (30%):** Quality and thoroughness of the documentation and presentation.

This project integrates machine learning, data analytics, and sports science into a practical application, emphasizing collaboration and clear communication through GitHub.

Deliverables:

1. Enter the GitHub information by March 17. The last time to update your code is March 17, 11:59 pm.

<https://docs.google.com/forms/d/e/1FAIpQLSeQzWiWaNdm8Ve90KZosVYVJ6TH8Yz76hWurOeFI46fJ2FDkw/viewform?usp=sharing> 
(<https://docs.google.com/forms/d/e/1FAIpQLSeQzWiWaNdm8Ve90KZosVYVJ6TH8Yz76hWurOeFI46fJ2FDkw/viewform?usp=sharing>)

2. Submit your presentation slides in ppt, pptx or pdf format by March 19th, 11:59 am. During your presentation you will be asked to walk through your GitHub page too.